C

IB MATH: APPLICATIONS & INTERPRETATION • UNIT 3.01

Gradient Descent & Optimization

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 C	Q3 C	Q4 C	Q5 A
Q6 A	Q7 A	Q8 A	Q9 B	Q10 D
Q11 D	Q12 C	Q13 C	Q14 A	Q15 A
Q16 C	Q17 C	Q18 D	Q19 B	Q20 A
Q21 B	Q22 C	Q23 C	Q24 A	Q25 A

Gradient Descent & Optimization Practice Exercises

1 MCQ

In gradient descent, a weight currently has value $w = 20$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 8$ and the learning rate is $\eta = 0.2$, find the updated weight w'.

A 12.00

B 21.60

C 18.40 ✓

D 19.80

WORKED SOLUTION & VERIFICATION

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 20 - (0.2)(8) = 20 - 1.60 = 18.40$.

2 MCQ

In gradient descent, a weight currently has value $w = 30$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 12$ and the learning rate is $\eta = 0.3$, find the updated weight w'.

A 18.00

B 33.60

C 26.40 ✓

D 29.70

WORKED SOLUTION & VERIFICATION

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 30 - (0.3)(12) = 30 - 3.60 = 26.40$.

3 MCQ

In gradient descent, a weight currently has value $w = 60$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 24$ and the learning rate is $\eta = 0.6$, find the updated weight w'.

A 36.00

B 74.40

C 45.60 ✓

D 59.40

WORKED SOLUTION & VERIFICATION

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 60 - (0.6)(24) = 60 - 14.40 = 45.60$.

4 MCQ

Consider the scalar loss function $L(w) = 2w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 2$.

- A $\nabla L(2) = 16$
- B $\nabla L(2) = 8$
- C $\nabla L(2) = 0$ ✓
- D $\nabla L(2) = -4$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(2)w - 8 = 4w - 8$. At $w = 2$: $\nabla L = 4(2) - 8 = 8 - 8 = 0$.

5 MCQ

Consider the scalar loss function $L(w) = 3w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 4$.

- A $\nabla L(4) = 16$ ✓
- B $\nabla L(4) = 24$
- C $\nabla L(4) = 32$
- D $\nabla L(4) = 4$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(3)w - 8 = 6w - 8$. At $w = 4$: $\nabla L = 6(4) - 8 = 24 - 8 = 16$.

6 MCQ

Consider the scalar loss function $L(w) = 4w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 6$.

- A $\nabla L(6) = 40$ ✓
- B $\nabla L(6) = 48$
- C $\nabla L(6) = 56$
- D $\nabla L(6) = 16$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(4)w - 8 = 8w - 8$. At $w = 6$: $\nabla L = 8(6) - 8 = 48 - 8 = 40$.

7 MCQ

Consider the scalar loss function $L(w) = 5w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 8$.

- A $\nabla L(8) = 72$ ✓
- B $\nabla L(8) = 80$
- C $\nabla L(8) = 88$
- D $\nabla L(8) = 32$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(5)w - 8 = 10w - 8$. At $w = 8$: $\nabla L = 10(8) - 8 = 80 - 8 = 72$.

8 MCQ

Consider the scalar loss function $L(w) = 6w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 10$.

- **A** $\nabla L(10) = 112$ ✓
- **B** $\nabla L(10) = 120$
- **C** $\nabla L(10) = 128$
- **D** $\nabla L(10) = 52$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(6)w - 8 = 12w - 8$. At $w = 10$: $\nabla L = 12(10) - 8 = 120 - 8 = 112$.

9 MCQ

Consider the quadratic loss function $L(w) = w^2 - 4w + 3$. Find the value of w where the function reaches its minimum.

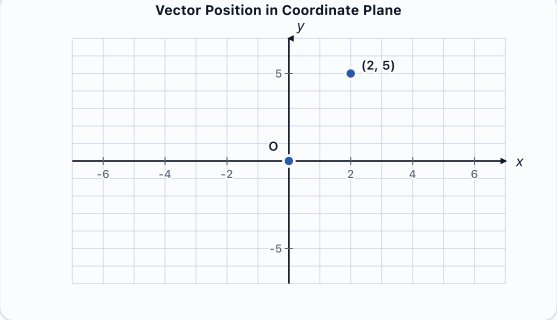
- **A** $w = 3$
- **B** $w = 2$ ✓
- **C** $w = 4$
- **D** $w = 1$

WORKED SOLUTION & VERIFICATION

The given loss function is a quadratic function $L(w) = w^2 - 4w + 3$. This represents a parabola opening upwards. The minimum value of a quadratic function of the form $aw^2 + bw + c$ occurs at $w = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -4$. Therefore, the minimum occurs at $w = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. Alternatively, we can find the minimum by taking the derivative and setting it to zero: $L'(w) = 2w - 4$. Setting $L'(w) = 0$ gives $2w - 4 = 0$, which implies $2w = 4$, so $w = 2$.

10 MCQ

Consider a univariate regression model where the predicted output $\hat{y}$ is given by $\hat{y} = ax + b$. The loss function is the Mean Squared Error (MSE), defined as $L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. For a single data point $(x_1, y_1) = (2, 5)$, the loss is $L = (5 - (2a + b))^2$. The gradient of the loss function with respect to a is $\frac{\partial L}{\partial a} = -2(5 - (2a + b)) \cdot 2$. If during gradient descent, with a learning rate of $\eta = 0.1$, the parameters a and b are updated from $a_{old} = 1$ and $b_{old} = 1$, to $a_{new} = 0.8$ and $b_{new} = 0.8$, what is the value of b that would have resulted in $\frac{\partial L}{\partial b} = 0$ for this single data point, assuming $a = 1$ and $b = 1$ were the values *before* the gradient calculation that led to the update to $a_{new} = 0.8, b_{new} = 0.8$?



A 3

B 4

C 6

D 5 ✓

WORKED SOLUTION & VERIFICATION

The loss function for a single data point is $L = (y_1 - (ax_1 + b))^2$. Given $(x_1, y_1) = (2, 5)$, so $L = (5 - (2a + b))^2$. The gradient of the loss with respect to b is $\frac{\partial L}{\partial b} = 2(5 - (2a + b)) \cdot (-1) = -2(5 - 2a - b)$. We are asked to find the value of b for which $\frac{\partial L}{\partial b} = 0$, assuming $a = 1$ and $b = 1$ were the parameters *before* the update step that led to $a_{new} = 0.8, b_{new} = 0.8$. This means we evaluate the gradient at $a = 1$ and the *unknown* value of b. Setting $\frac{\partial L}{\partial b} = 0$: $-2(5 - 2(1) - b) = 0 5 - 2 - b = 0 3 - b = 0 b = 3$. The provided gradient update information $\frac{\partial L}{\partial a} = -2(5 - (2a + b)) \cdot 2$ and the learning rate $\eta = 0.1$ are distractors. The question specifically asks for the value of b that makes $\frac{\partial L}{\partial b} = 0$ *at the point where $a = 1$*. The update rule to $a_{new} = 0.8, b_{new} = 0.8$ implies that the gradient of a was $\frac{a_{old} - a_{new}}{\eta} = \frac{1 - 0.8}{0.1} = 2$, and the gradient of b was $\frac{b_{old} - b_{new}}{\eta} = \frac{1 - 0.8}{0.1} = 2$. Let's verify: If $a = 1$ and $b = 3$, then $\frac{\partial L}{\partial b} = -2(5 - 2(1) - 3) = -2(5 - 2 - 3) = -2(0) = 0$. If $a = 1$ and $b = 3$, $\frac{\partial L}{\partial a} = -2(5 - 2(1) - 3) \cdot 2 = -2(0) \cdot 2 = 0$. This scenario implies that the point $a = 1, b = 3$ is a minimum for the loss function, but the question is about finding the value of b that makes the gradient of b zero at $a = 1$, which is $b = 3$. The provided update $a_{new} = 0.8, b_{new} = 0.8$ from $a_{old} = 1, b_{old} = 1$ is a different step in the optimization process and is not directly used to find the value of b that makes the gradient of b zero at $a = 1$.

11 MCQ

Considering gradient descent, a weight currently has value $w = 30$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 12$ and the learning rate is $\eta = 0.3$, find the updated weight w'.

A 18.00

B 33.60

C 29.70

D 26.40 ✓

WORKED SOLUTION & VERIFICATION

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 30 - (0.3)(12) = 30 - 3.60 = 26.40$.

12 MCQ

Examining gradient descent, a weight currently has value $w = 60$. If the gradient of the loss is $\frac{\partial L}{\partial w} = 24$ and the learning rate is $\eta = 0.6$, find the updated weight w'.

A 59.40

B 36.00

C 45.60 ✓

D 74.40

WORKED SOLUTION & VERIFICATION

Using the gradient descent update rule: $w' = w - \eta \frac{\partial L}{\partial w} = 60 - (0.6)(24) = 60 - 14.40 = 45.60$.

13 MCQ

Consider the following quadratic loss function for a regression problem: $L(y, \hat{y}) = (y - \hat{y})^2$. If the predicted value $\hat{y} = 5$ and the true value $y = 3$, what is the value of the loss function?

A -2

B 9

C 4 ✓

D 2

WORKED SOLUTION & VERIFICATION

The quadratic loss function is defined as $L(y, \hat{y}) = (y - \hat{y})^2$. Given the true value $y = 3$ and the predicted value $\hat{y} = 5$, we substitute these values into the loss function:

$L(3, 5) = (3 - 5)^2$

$L(3, 5) = (-2)^2$

$L(3, 5) = 4$

Therefore, the value of the loss function is 4.

14 MCQ

An Ordinary Least Squares linear model $\hat{f}(x) = 23x + 14$ is evaluated at $x = 19$. Determine the predicted response $\hat{f}(19)$.

A **451** ✓

B 455

C 474

D 437

WORKED SOLUTION & VERIFICATION

Evaluating $23(19) + 14 = 451$.

15 MCQ

Consider the following quadratic loss function: $L(w) = 3w^2 - 12w + 10$. What is the value of w that minimizes this loss function?

- **A** 2 ✓
- **B** 12
- **C** 1
- **D** 4

WORKED SOLUTION & VERIFICATION

To find the minimum of the quadratic loss function $L(w) = 3w^2 - 12w + 10$, we can find the derivative with respect to w and set it to zero. The derivative of $L(w)$ with respect to w is: $\frac{dL}{dw} = 6w - 12$. Setting the derivative to zero to find the critical point: $6w - 12 = 0$ $6w = 12$ $w = \frac{12}{6}$ $w = 2$. The second derivative is $\frac{d^2L}{dw^2} = 6$, which is positive, confirming that $w = 2$ corresponds to a minimum.

16 MCQ

Consider the scalar loss function $L(w) = 7w^2 - 8w + 5$. Find the analytical gradient $\nabla L(w)$ evaluated at $w = 12$.

- **A** $\nabla L(12) = 168$
- **B** $\nabla L(12) = 176$
- **C** $\nabla L(12) = 160$ ✓
- **D** $\nabla L(12) = 76$

WORKED SOLUTION & VERIFICATION

Differentiate with respect to w: $\frac{dL}{dw} = 2(7)w - 8 = 14w - 8$. At $w = 12$: $\nabla L = 14(12) - 8 = 168 - 8 = 160$.

17 MCQ

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

- **A** $\nabla L = [10, -8]^T$
- **B** $\nabla L = [8, 6]^T$
- **C** $\nabla L = [8, -6]^T$ ✓
- **D** $\nabla L = [4, -3]^T$

WORKED SOLUTION & VERIFICATION

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(2)w_1 = 4(2) = 8$; $\frac{\partial L}{\partial w_2} = 2(3)w_2 = 6(-1) = -6$. Therefore, $\nabla L = [8, -6]^T$.

18 MCQ

A 2D loss surface is defined as $L(w_1, w_2) = 3w_1^2 + 5w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

- **A** $\nabla L = [14, -12]^T$
- **B** $\nabla L = [12, 10]^T$
- **C** $\nabla L = [6, -5]^T$
- **D** $\nabla L = [12, -10]^T$ ✓

WORKED SOLUTION & VERIFICATION

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(3)w_1 = 6(2) = 12$; $\frac{\partial L}{\partial w_2} = 2(5)w_2 = 10(-1) = -10$. Therefore, $\nabla L = [12, -10]^T$.

19 MCQ

In an applied problem, a 2D loss surface is defined as $L(w_1, w_2) = 4w_1^2 + 7w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [18, -16]^T$

B $\nabla L = [16, -14]^T$ ✓

C $\nabla L = [16, 14]^T$

D $\nabla L = [8, -7]^T$

WORKED SOLUTION & VERIFICATION

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(4)w_1 = 8(2) = 16$; $\frac{\partial L}{\partial w_2} = 2(7)w_2 = 14(-1) = -14$. Therefore, $\nabla L = [16, -14]^T$.

20 MCQ

Suppose a 2D loss surface is defined as $L(w_1, w_2) = 5w_1^2 + 9w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [20, -18]^T$ ✓

B $\nabla L = [22, -20]^T$

C $\nabla L = [20, 18]^T$

D $\nabla L = [10, -9]^T$

WORKED SOLUTION & VERIFICATION

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(5)w_1 = 10(2) = 20$; $\frac{\partial L}{\partial w_2} = 2(9)w_2 = 18(-1) = -18$. Therefore, $\nabla L = [20, -18]^T$.

21 MCQ

An observed 2D loss surface is defined as $L(w_1, w_2) = 6w_1^2 + 11w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [26, -24]^T$

B $\nabla L = [24, -22]^T$ ✓

C $\nabla L = [24, 22]^T$

D $\nabla L = [12, -11]^T$

WORKED SOLUTION & VERIFICATION

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(6)w_1 = 12(2) = 24$; $\frac{\partial L}{\partial w_2} = 2(11)w_2 = 22(-1) = -22$. Therefore, $\nabla L = [24, -22]^T$.

22 MCQ

Consider a 2D loss surface is defined as $L(w_1, w_2) = 7w_1^2 + 13w_2^2$. Find the gradient vector ∇L at the point $(w_1, w_2) = (2, -1)$.

A $\nabla L = [30, -28]^T$

B $\nabla L = [28, 26]^T$

C $\nabla L = [28, -26]^T$ ✓

D $\nabla L = [14, -13]^T$

WORKED SOLUTION & VERIFICATION

Compute partial derivatives: $\frac{\partial L}{\partial w_1} = 2(7)w_1 = 14(2) = 28$; $\frac{\partial L}{\partial w_2} = 2(13)w_2 = 26(-1) = -26$. Therefore, $\nabla L = [28, -26]^T$.

23 MCQ

Consider minimizing the convex loss $L(w) = (w + 3)^2$ (Run #1) using gradient descent with learning rate $\eta = 0.25$ starting at $w_0 = -1$. Compute $w_1 = w_0 - \eta L'(w_0)$.

A -3.50	B 0.00
C -2 ✓	D 4

WORKED SOLUTION & VERIFICATION

Here $L'(w) = 2(w - (-3))$, so $L'(-1) = 4$. Updating gives $w_1 = -1 - 0.25(4) = -2$.

24 MCQ

Consider minimizing the convex loss $L(w) = 2(w + 2)^2$ (Run #2) using gradient descent with learning rate $\eta = 0.25$ starting at $w_0 = 1$. Compute $w_1 = w_0 - \eta L'(w_0)$.

A -2 ✓	B -3.50
C 4.00	D 12

WORKED SOLUTION & VERIFICATION

Here $L'(w) = 4(w - (-2))$, so $L'(1) = 12$. Updating gives $w_1 = 1 - 0.25(12) = -2$.

25 MCQ

Consider minimizing the convex loss $L(w) = 3(w + 1)^2$ (Run #3) using gradient descent with learning rate $\eta = 0.25$ starting at $w_0 = 3$. Compute $w_1 = w_0 - \eta L'(w_0)$.

A -3 ✓	B 9.00
C 24	D -4.50

WORKED SOLUTION & VERIFICATION

Here $L'(w) = 6(w - (-1))$, so $L'(3) = 24$. Updating gives $w_1 = 3 - 0.25(24) = -3$.